(i)

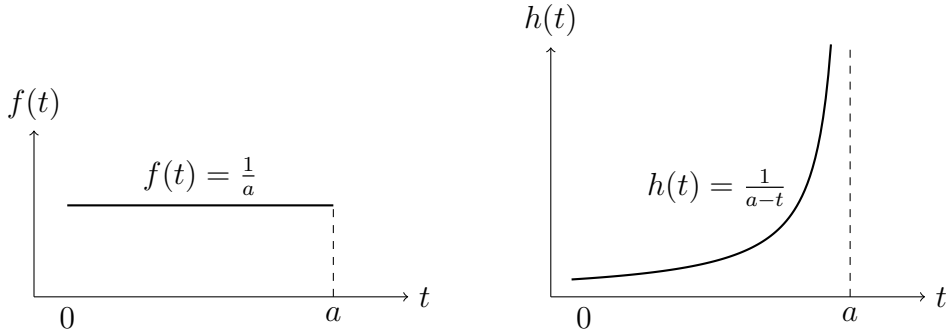
$$\begin{aligned} P(t \le T < t+\delta t \mid T \ge t) &= \frac{P(t \le T < t+\delta t)}{P(T \ge t)} \\ &= \frac{F(t+\delta t) - F(t)}{1 - F(t)} \\ &\approx \frac{f(t)\delta t}{1 - F(t)} \\ &= h(t)\delta t \end{aligned}$$

(ii)

$$F(t) = \frac{t}{a}, \qquad 0 < t < a$$

$$f(t) = F'(t) = \frac{1}{a}$$

$$h(t) = \frac{f(t)}{1 - F(t)} = \frac{1/a}{1 - \frac{t}{a}} = \boxed{\frac{1}{a-t}}$$



(iii)

$$\begin{aligned} h(t) = \frac{f(t)}{1 - F(t)} &= \frac{1}{t} \\ \frac{F'(t)}{1 - F(t)} &= \frac{1}{t} \\ -\ln(1 - F(t)) &= \ln t + C \\ 1 - F(t) &= \frac{C}{t} \end{aligned}$$

$\because T > a,\ F(a) = 0 \implies C = a$

$$\implies F(t) = 1 - \frac{a}{t}$$

$$\therefore \boxed{f(t) = F'(t) = \frac{a}{t^2}, \qquad t > a}$$

(iv) When $h(t) = k, \qquad t > b$

$$\frac{F'(t)}{1 - F(t)} = k$$
$$-\ln(1 - F(t)) = kt + C$$
$$1 - F(t) = Ce^{-kt}$$

$\because h(t) = 0$ for $t \leq b, \quad F(b) = 0$

$$1 = Ce^{-kb} \implies C = e^{kb}$$

$$1 - F(t) = e^{-k(t-b)}$$

$$F(t) = 1 - e^{-k(t-b)}$$

$$\therefore f(t) = F'(t) = ke^{-k(t-b)}, \qquad t > b$$

$$\therefore h(t) = \frac{ke^{-k(t-b)}}{e^{-k(t-b)}} = k$$

$$\boxed{h(t) \text{ is constant for } t > b \iff f(t) = ke^{-k(t-b)}}$$

(v)

$$h(t) = \left(\frac{\lambda}{\theta^\lambda}\right) t^{\lambda-1}$$

$$\frac{F'(t)}{1 - F(t)} = \frac{\lambda}{\theta^\lambda} t^{\lambda-1}$$
$$-\ln(1 - F(t)) = \int_0^t \frac{\lambda}{\theta^\lambda} u^{\lambda-1}\, du$$
$$-\ln(1 - F(t)) = \frac{t^\lambda}{\theta^\lambda}$$
$$1 - F(t) = e^{-(t/\theta)^\lambda}$$

$$\therefore f(t) = h(t)(1 - F(t))$$

$$\boxed{f(t) = \frac{\lambda}{\theta^{\lambda}} t^{\lambda-1} e^{-(t/\theta)^{\lambda}}, \qquad t > 0}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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